

**“SafeLAN”: Adaptive Secure File System for LAN
environment**

25-26J-067

Project Proposal Report

HIRIMUTHUGODA N. A.

B.Sc.(Hons) Degree in Information technology specializing in Cyber
Security

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology

August 2025

**“SafeLAN”: Adaptive Secure File System for LAN
environment**

25-26J-067

Project Proposal Report

HIRIMUTHUGODA N. A.

B.Sc.(Hons) Degree in Information technology specializing in Cyber
Security

Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology

August 2025

DECLARATION OF CANDIDATE AND SUPERVISOR

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

HIRIMUTHUGODA N. A.



.....

Date

28/08/2025

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor



.....

Date

28/08/2025

Mr. Kanishka Yapa

ABSTRACT

Concerns about safe access to private data in local area network (LAN) settings have increased as a result of the digitisation of workplace operations. Vulnerabilities like phishing, spoofing, deepfakes, and privacy risks related to permanent biometric storage are frequently revealed by existing authentication methods like passwords, one-time codes, or conventional biometrics. This study suggests SafeLAN, a gesture-based, code-driven biometric authentication system combined with a lockdown browser, as a solution to these issues. According to the suggested framework, a user's mobile device receives a temporary numeric code that needs to be replicated using hand gestures recorded by a webcam and validated in real time using a lightweight recognition model. SafeLAN, in contrast to traditional methods, protects privacy by removing permanent biometric storage, improves resistance to spoofing and phishing by using dual verification, and stops unwanted data access by limiting file operations in a secure lockdown environment. System design, gesture recognition model training, and thorough evaluation in LAN-based environments are all part of the research methodology, which focusses on user experience, response time, and authentication accuracy. By developing privacy-conscious authentication frameworks for sensitive digital environments, a workable, hardware-independent solution that enhances security and usability is anticipated, with important ramifications for businesses, academia, and society.

Keywords: biometric authentication, gesture recognition, privacy-preserving security, LAN security, lockdown browser, phishing resistance, spoofing protection

TABLE OF CONTENT

DECLARATION OF CANDIDATE AND SUPERVISOR	iii
ABSTRACT	iv
TABLE OF CONTENT	v
1 List of Figures	vi
2 INTRODUCTION	7
2.1 Background and Literature Survey	8
2.2 Research Gap	10
2.3 Research Problem	11
3 OBJECTIVES	14
3.1 Main Objectives	14
3.2 Specific Objectives	14
4 METHODOLOGY	16
4.1 System Diagram and Component Explanation	16
4.2 Implementation Approach.....	17
4.3 Required Tools and Resources	17
4.4 Data Collection and Training	17
4.5 Project Timeline	18
4.6 Expected Outcome	18
5 PROJECT REQUIREMENTS	19
5.1 Functional Requirements	19
5.2 Non-Functional Requirements	20

5.3	Expected Test Cases	21
5.4	User Requirements	22
5.5	System Requirements.....	23
5.5.1	Software Requirements	23
5.5.2	Hardware Requirements.....	23
6	DESCRIPTION OF PERSONAL AND FACILITIES	24
7	BUDGET AND BUDGET JUSTIFICATION	25
8	REFERENCES	26

1 List of Figures

Table 1: Implementation description.....	17
Table 2: Project Timeline	18
Table 3: Budget Allocation	25

2 INTRODUCTION

Secure access in local network (LAN) environments has become a concern as office operations and internal file handling become more digitalised. Sensitive information that is shared between teams or departments is vulnerable to credential theft, insider threats, and unauthorised access, particularly in industries like finance, law, or research. Passwords and single-use codes are examples of traditional methods that only authenticate users once, leaving sensitive files vulnerable during or after login sessions.

Modern secure systems prioritise multi-layered, context-aware access control to fill these gaps, especially when opening or working with important files. Even though conventional biometrics like fingerprint or facial recognition work well, they frequently have privacy risks, need specialised hardware, or present usability issues in shared desktop settings.

This study investigates a new gesture-based biometric system that is integrated into a lockdown browser to provide phishing-resistant and privacy-preserving file access. A random number code is sent to the user's phone when they open a file via SafeLAN. Following that, the user has to replicate this code using hand gestures (like counting fingers), which are recorded by the device's camera and validated by gesture recognition models.

2.1 Background and Literature Survey

Securely sharing and accessing sensitive files is a recurring problem in contemporary work environments, particularly those that operate within closed local area networks (LANs). Even though businesses spend a lot of money on access control and endpoint security, internal systems frequently rely on antiquated techniques like static passwords, email-based links, or one-time codes. These methods are more susceptible to brute-force login attempts, phishing scams, and insider threats, especially when user behaviour is predictable or not closely watched.

In order to address these issues, biometric authentication has emerged as a crucial field of study. While facial recognition, iris scanning, and fingerprint scanning have shown promise, they also pose serious risks. These include hardware dependencies, privacy violations, and the possibility of irreversible identity theft in the event that biometric data is compromised. Recent developments have begun to investigate gesture-based recognition and behavioural biometrics as privacy-preserving and hard-to-spoof substitutes.

Despite developments, few systems have concentrated on real-time authentication logic in conjunction with gesture-based biometrics, particularly in controlled environments such as lockdown browsers. The existing literature is lacking in this area. Although gesture recognition has been applied to accessibility interfaces, gaming, and education, little is known about how to incorporate it into security-sensitive applications like file access control. This research aims to contribute to the field by proposing a novel gesture-based biometric mechanism within SafeLAN, a secure file-sharing system. The proposed method introduces a two-step process:

1. A random numeric code is sent to the user's mobile device.
2. The user reproduces the code using hand gestures, which are verified via camera-based recognition.

This approach ensures that access is granted only within a lockdown browser and only to the intended recipient, without exposing permanent biometric data or relying on phishing-prone login methods.

The significance of this research lies in its practical impact:

- For organizations and funders, it reduces the risk of internal data leaks and cyberattacks on LAN systems.
- For the academic and scientific community, it opens a new path in gesture-based, temporary biometric authentication systems.
- For society, it proposes a privacy-conscious and user-friendly way of securing access to digital information in corporate and institutional settings.

By improving both security and usability, this work contributes to the broader goal of creating resilient authentication frameworks suitable for the evolving threat landscape.

2.2 Research Gap

Current biometric authentication systems primarily rely on physical biometrics (such as fingerprints and iris patterns) or behavioral biometrics (including keystrokes and voice recognition). [1] There are certain privacy requirements when dealing with biometric authentication

1. Irreversibility: Biometric systems must ensure that it is computationally infeasible to reconstruct the original biometric samples from stored or processed data. Without this, compromised templates can lead to permanent identity theft. [2]
2. Unlinkability: It is essential to prevent the association of a user's biometric data across different systems. Linking biometric records can compromise user privacy and enable profiling across platforms. [2]
3. Soft-Biometric Privacy: Systems should prevent the extraction of unintended soft-biometric attributes, such as gender, ethnicity, or age, particularly when such information is not required for the intended authentication purpose. [2]

While current solutions such as cancellable biometrics (storing modified, non-reversible versions) [4], biometric cryptosystems (binding or generating cryptographic keys from biometrics) [3], homomorphic encryption (performing matching on encrypted data without revealing raw biometrics) [5], soft-biometric minimization (removing attributes like age, gender, or race) [3] and soft-biometric protection (masking sensitive attributes) [6] address some of these issues, they still face limitations. These include the need to store sensitive data, reliance on complex computations, and dependence on specialized hardware.

There remains a clear gap for authentication methods that are:

- Privacy-preserving, without storing permanent biometric templates.
- Hardware-independent, allowing deployment in diverse environments.
- Resistant to phishing, spoofing, and brute-force attacks.

The proposed research addresses this gap by introducing a gesture-based biometric authentication system that:

- Eliminates permanent storage of biometric data, reducing privacy concerns and database management overhead.
- Integrates seamlessly with a lockdown browser, ensuring files are accessed securely.
- Combines code-based verification with gesture recognition, providing protection against phishing and brute-force attacks.

2.3 Research Problem

Even with advances in biometric security, sensitive file access in LAN environments continues to face several critical threats:

1. Spoofing Attacks: Attackers can fake biometric samples to impersonate legitimate users and gain unauthorized access. Current mitigations include two-factor authentication (2FA) and liveness detection, but these measures are not always foolproof. [7]

2. Data can be intercepted or accessed without authorization during storage or transmission. Strong encryption and secure communication protocols help, but the permanent storage of sensitive biometric data still poses risks. [8]
3. Data Corruption: Biometric templates can be manipulated or corrupted, compromising authentication integrity. Solutions such as data integrity checks and version control help mitigate this, but data storing is costly [9]
4. Deepfakes: AI-generated videos or audio can mimic users' faces or voices, bypassing traditional authentication systems. AI-based deepfake detection and liveness verification are used to counter this threat. [10]
5. Phishing Attacks: Users may be tricked into providing biometric information on malicious websites. Security awareness training helps reduce this risk, but phishing remains a persistent challenge and training isn't very feasible everywhere. [11]

Research Question:

How can a gesture-based, code-driven biometric authentication system secure file access in LAN environments while addressing threats to existing biometric authentication mechanisms.

Research Goal:

The goal is to design and evaluate a system that ensures:

- Files open only via a lockdown browser.
- Users verify identity using temporary numeric codes reproduced as hand gestures, verified in real-time.

- No permanent biometric data is stored, reducing privacy risks and database management costs.
- The system is resilient against spoofing, phishing, deepfakes, data corruption, and unauthorized data access, improving overall security.

By tackling these challenges, this research givesw a practical solution for secure file access in LAN environments and contributes to the development of privacy-preserving biometric authentication methods.

3 OBJECTIVES

3.1 Main Objectives

To create a gesture-based, code-driven biometric authentication system that protects against common security threats like spoofing, phishing, and brute-force attacks, ensures privacy preservation, and allows safe file access in LAN environments.

3.2 Specific Objectives

1. Implement a lockdown browser environment to ensure files are accessed only in a secure and controlled setting.
2. Design a gesture-based verification mechanism where temporary numeric codes are reproduced through hand gestures and verified in real-time.
3. Minimize privacy risks by avoiding permanent storage of biometric data and sensitive user information.
4. Enhance security against common threats, including spoofing, deepfakes, phishing attacks, and data corruption.
5. Evaluate system effectiveness and usability by testing authentication accuracy, response time, and user experience in LAN environments.

4 METHODOLOGY

We suggest an organised development and testing methodology that integrates secure system integration, gesture recognition, and biometric authentication to meet the goals of this study. System design, component implementation, testing, and evaluation are among the distinct phases that make up the entire process.

4.1 System Diagram and Component Explanation

User Receives Code → User Performs Hand Gesture → Gesture Captured via Webcam one by one → Gesture Recognition Engine → Code Match Verification → Access Granted in Lockdown Browser

Components Involved:

1. **Code Generator:** Random numeric code is generated and sent to the authenticated user's mobile device.
2. **Gesture Interface:** User is prompted to reproduce the code using hand gestures.
3. **Camera Module:** Standard webcam captures the gesture in real time.
4. **Gesture Recognition Engine:** A lightweight machine learning model processes the hand gesture and interprets the numeric values.
5. **Verification Module:** Compares the input gesture to the original code sent to the user.
6. **Lockdown Browser Module:** If the match is successful, the requested file opens within a secure lockdown browser window.

4.2 Implementation Approach

The project will be carried out in the following phases:

Phase	Description
1. System Design	Design the overall system architecture, define module interactions, and sketch UI wireframes.
2. Code Delivery System	Implement secure code generation and delivery via SMS/email.
3. Gesture Recognition Engine	Train/test hand gesture recognition model using a computer vision library (e.g., MediaPipe/OpenCV).
4. Lockdown Browser Setup	Integrate a secure file viewer that restricts copying, printing, or screenshotting.
5. Integration & Testing	Combine all modules and run security and usability tests.
6. Evaluation	Measure authentication accuracy, false rejection/acceptance rates, and user satisfaction.

Table 1: Implementation description

4.3 Required Tools and Resources

- **Hardware:** Laptop/PC with a webcam.
- **Software:** Python, OpenCV, MediaPipe, lockdown browser framework.
- **Other Resources:** Internet for data collection and model training datasets; testing environment within a LAN.

4.4 Data Collection and Training

To build the gesture recognition engine:

- A sample dataset of hand gestures representing digits (0–9) will be collected using a webcam.
- Each gesture will be recorded multiple times under varying lighting conditions and angles.
- The data will be labeled and used to train a basic classification model

4.5 Project Timeline

Task	Duration	Deadline
Requirement Analysis & Design	1 Month	Month 1
Code Generator & Delivery	1 Month	Month 2
Gesture Recognition Development	1.5 Months	Month 3.5
Lockdown Browser Integration	2 Weeks	Month 4
System Integration & Testing	1 Month	Month 5
Final Evaluation & Report Writing	1 Month	Month 6

Table 2: Project Timeline

4.6 Expected Outcome

The anticipated outcome is a secure, privacy-preserving biometric authentication system that:

- Does not store permanent biometric data.
- It can be easily used in LAN-based environments without the need for extra hardware.
- Prevents spoofing, phishing, and unauthorized access to sensitive files.

This solution can be deployed in office settings, educational institutions, or healthcare LANs for controlled document access.

5 PROJECT REQUIREMENTS

5.1 Functional Requirements

The system must perform the following core functions:

- The system must generate a unique numeric code for each authentication attempt.
- The system must send the generated code to the user's registered mobile device.
- The system must allow users to input the received code through hand gestures.
- The system must capture the user's hand gestures in real time using a webcam.
- The system must accurately recognize numeric digits from the captured hand gestures.
- The system must compare the recognized gesture code with the originally sent code.
- The system must allow access only if the code from the gesture matches the sent code.
- The system must open files only within a lockdown browser after successful authentication.
- The system must restrict file copying, printing, and screenshots within the lockdown browser.
- The system must invalidate codes after a predefined timeout period.
- The system must display a clear error message if gesture recognition or matching code fails.
- The system must allow users to retry authentication if an error occurs.

5.2 Non-Functional Requirements

The system must also meet the following quality requirements:

- The system must provide accurate gesture recognition with a minimum 90% success rate.
- The system must run on standard hardware without requiring specialized biometric sensors.
- The system interface must be user-friendly and intuitive for non-technical users.
- The system must be compatible with commonly used operating systems (e.g., Windows, Linux).
- The gesture recognition model must be lightweight and optimized for real-time performance.
- The system must ensure reliable operation in low-light or varied lighting conditions.
- The system must be scalable for use by multiple users within a LAN environment.
- The system must recover gracefully from errors without crashing or freezing.
- The system must operate consistently across different webcam resolutions.
- The system must consume minimal system resources during authentication.

5.3 Expected Test Cases

The system will be tested under different scenarios to validate correctness and robustness:

Test Case 1: Valid Gesture Input

Input: User receives a valid code and correctly performs the gesture.

Expected Result: The system recognizes the gesture, verifies the code, and grants access via lockdown browser.

Test Case 2: Invalid Gesture Input

Input: User performs incorrect hand gestures that don't match the code.

Expected Result: Access is denied with an appropriate error message.

Test Case 3: Expired Code Usage

Input: User attempts to authenticate with a code that has expired.

Expected Result: System displays a timeout error and requests a new code.

Test Case 4: Gesture Recognition in Low Light

Input: User performs the correct gesture under poor lighting conditions.

Expected Result: The system still recognizes the gesture accurately or prompts user to move to a well-lit area.

Test Case 5: Code Sent but Gesture Not Performed

Input: Code is sent but the user doesn't perform the gesture within the allowed time.

Expected Result: System invalidates the session and prompts the user to request a new code.

Test Case 6: Gesture Performed in Multiple Angles

Input: User performs the gesture from different angles and distances.

Expected Result: System consistently recognizes the gesture across variations.

5.4 User Requirements

From the end-user perspective.....

- The user must be able to receive a unique numeric code for each authentication session.
- The user must be able to input the received code using hand gestures.
- The user must be able to perform gestures in front of a webcam without needing any specialized equipment.
- The user must be able to view error messages and retry the authentication if a mistake occurs.
- The user must access files only through a secure lockdown browser after successful authentication.
- The user must be notified when the authentication code has expired or is invalid.
- The user must experience fast and smooth authentication with minimal delay.

5.5 System Requirements

5.5.1 Software Requirements

- Operating System: Windows 10 or later / Linux (Ubuntu 20.04 or later)
- Python 3.8 or higher (for gesture recognition and backend logic)
- OpenCV and MediaPipe libraries (for hand gesture detection)
- Web browser with lockdown capabilities (e.g., SEB - Safe Exam Browser or custom wrapper)
- Local web server or backend framework (e.g., Flask/Django) for managing authentication logic and file access

5.5.2 Hardware Requirements

- Standard PC or laptop with at least Intel i5 processor or equivalent
- Minimum 8 GB RAM for smooth gesture recognition and system operation
- Webcam (HD 720p or higher) for capturing hand gestures

6 DESCRIPTION OF PERSONAL AND FACILITIES

The following individuals act as facilitators to help the study objectives be met:

|Mr. Kanishka Yapa : Sri Lanka Institute of Information Technology (SLIIT)

Mr. Deemantha Siriwardena : Sri Lanka Institute of Information Technology (SLIIT)

7 BUDGET AND BUDGET JUSTIFICATION

Budget:

Item	Estimated Cost (LKR)
Standard PC/Laptop	200,000
Total	200,000

Table 3: Budget Allocation

Budget Justification:

- PC/Laptop: Needed to run gesture recognition algorithms, backend services, and test environments.

This budget ensures that the project has the necessary hardware and software resources to implement, test, and evaluate a functional gesture-based biometric authentication system in a LAN environment.

8 REFERENCES

- [1] K. Krishna Prakasha and U. Sumalatha, "Privacy-Preserving Techniques in Biometric Systems: Approaches and Challenges," *IEEE Access*, vol. 13, pp. 32584–32616, 2025, doi: <https://doi.org/10.1109/access.2025.3541649>.
- [2] Z. Rui and Z. Yan, "A Survey on Biometric Authentication: Toward Secure and Privacy-Preserving Identification," *IEEE Access*, vol. 7, pp. 5994–6009, 2019, doi: <https://doi.org/10.1109/access.2018.2889996>.
- [3] Pietro Melzi, C. Rathgeb, Rubén Tolosana, Rubén Vera-Rodríguez, and C. Busch, "An Overview of Privacy-enhancing Technologies in Biometric Recognition," *arXiv (Cornell University)*, Jun. 2022, doi: <https://doi.org/10.48550/arxiv.2206.10465>.
- [4] N. K. Ratha, J. H. Connell, and R. M. Bolle, "Enhancing security and privacy in biometrics-based authentication systems," *IBM Systems Journal*, vol. 40, no. 3, pp. 614–634, 2001, doi: <https://doi.org/10.1147/sj.403.0614>.
- [5] R. Arjona and I. Baturone, "A Post-Quantum Biometric Template Protection Scheme Based on Learning Parity With Noise (LPN) Commitments," *IEEE Access*, vol. 8, pp. 182355–182365, Jan. 2020, doi: <https://doi.org/10.1109/access.2020.3028703>.
- [6] P. Terhorst *et al.*, "PE-MIU: A Training-Free Privacy-Enhancing Face Recognition Approach Based on Minimum Information Units," *IEEE Access*, vol. 8, pp. 93635–93647, Jan. 2020, doi: <https://doi.org/10.1109/access.2020.2994960>.
- [7] Divine Senanu Ametefe *et al.*, "Enhancing Fingerprint Authentication: A Systematic Review of Liveness Detection Methods Against Presentation Attacks," *Journal of The Institution of Engineers (India) Series B*, May 2024, doi: <https://doi.org/10.1007/s40031-024-01066-3>.

- [8] E. D. Alalade, M. Mahyoub, and Ashraf Matrawy, "Privacy Engineering in Smart Home (Sh) Systems: A Comprehensive Privacy Threat Analysis and Risk Management Approach," Jan. 01, 2024.
https://www.researchgate.net/publication/378624253_Privacy_Engineering_in_Smart_Home_Sh_Systems_A_Comprehensive_Privacy_Threat_Analysis_and_Risk_Management_Approach
- [9] M. M. Charles and M. M. Mark, "Border Management Identification: The Biometric Technology to Detect Criminals and Terrorists Often Travel using Falsified Identity Documents," *Ssrn.com*, Aug. 26, 2024.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5006155
- [10] J. Jenkins and K. Roy, "Exploring deep convolutional generative adversarial networks (DCGAN) in biometric systems: a survey study," *Discover Artificial Intelligence*, vol. 4, no. 1, May 2024, doi:
<https://doi.org/10.1007/s44163-024-00138-z>.
- [11] A. Selwal, D. Sharma, M. Mann, S. Chakraborty, V. E. Balas, and O. E. Lieh, *Leveraging Computer Vision to Biometric Applications*. Boca Raton: Chapman and Hall/CRC, 2024. doi: <https://doi.org/10.1201/9781032614663>.

